

VPA IB DP CS — Topic 4: Computational Thinking (Weeks 10 – 18)

Topic 4: Computational Thinking 计算思维 Jìsuàn Sīwéi

IB Syllabus: Topic 4 (4.1 – 4.4) | Level: SL + HL | Duration: 9 weeks (30 hours)

Topic 4.1: Programming Fundamentals 编程基础 Biānchéng Jīchǔ

Week 10 — Pseudocode and Variables 伪代码与变量 Wěi Dàimǎ yǔ Biànlìang

Lesson	Content	Activities	Differentiation
1	IB pseudocode conventions; variables, constants, data types (integer, real, boolean, string, char) (伪代码, 变量, 常量)	Write pseudocode for simple calculations; identify errors	Bronze: Fill gaps; Silver: Write from scratch; Gold: Validate input
2	Operators: arithmetic (+, −, *, /, MOD, DIV), relational, logical	Evaluate expressions; predict outputs	Bronze: Integer arithmetic; Silver: Mixed types; Gold: Complex boolean
3	Selection: IF...THEN...ELSE...ENDIF; nested IF; CASE (选择结构, 条件语句)	Write grade calculator; letter classification	Bronze: Single IF; Silver: Nested; Gold: CASE with multiple conditions
4	Iteration: FOR, WHILE, REPEAT...UNTIL; loop control; infinite loops (循环, 迭代)	Sum of series; factorial; validate input with loop	Bronze: FOR loop; Silver: WHILE with condition; Gold: Nested loops
5	TOK: Is pseudocode a language? Does it have rules, meaning, and truth values like natural language?	Discussion; compare pseudocode to recipe, mathematical notation, poetry	All: Contribute; Gold: Reference philosophy of language

TOK Knowledge Question: Pseudocode occupies a space between natural language and formal programming languages. When we write pseudocode, are we creating a precise set of instructions, or are we merely suggesting an approach? Can pseudocode be "wrong," or is it too informal to have truth values?

Week 11 — Arrays and Strings 数组与字符串 Shùzǔ yǔ Zìfúchuàn

Lesson	Content	Activities
1	One-dimensional arrays: declaration, indexing, traversal (数组, 索引)	Store and display temperatures; calculate average
2	Two-dimensional arrays: rows, columns, nested loops for traversal	Matrix operations; game board representation

Lesson	Content	Activities
3	String operations: length, substring, concatenation, ASCII/Unicode comparison	Palindrome checker; password validator
4	File handling: open, read, write, close; CSV parsing (文件处理)	Read student data from file; calculate statistics
5	Formative: Programming exercises (3 problems, 45 minutes)	Timed coding; peer review

Week 12 — Sub-programs and Parameters 子程序与参数 Zǐchéngxù yǔ Cānshù

Lesson	Content	Activities
1	Procedures and functions: definition, call, return (过程, 函数)	Refactor code into functions; identify benefits
2	Parameters: value vs reference passing; local vs global scope	Trace programs with parameter passing; predict outputs
3	Built-in functions: mathematical, string, random; library usage	Documentation reading; use unfamiliar library
4	Modular design: cohesion, coupling, information hiding	Evaluate code modules for quality
5	IA Application: Plan modular structure for your IA project	Diagram modules; define interfaces

Topic 4.2: Data Structures 数据结构 Shùjù Jiégòu

Week 13 — Collections and Records 集合与记录 Jíhé yǔ Jìlù

Lesson	Content	Activities	Differentiation
1	Records: composite data types, fields, arrays of records (记录, 字段)	Define student record; array of students	Bronze: Define record; Silver: Array of records; Gold: Search by field
2	Lists/Collections: dynamic sizing, adding, removing, searching (集合, 列表)	Implement playlist manager; shopping list	Bronze: Add/remove; Silver: Search; Gold: Sort collection
3	Stacks: LIFO, push, pop, peek, isEmpty; applications (undo, back button) (栈)	Simulate stack with array; bracket checker	Bronze: Trace operations; Silver: Implement; Gold: Evaluate expressions
4	Queues: FIFO, enqueue, dequeue, circular queue; applications (print spool, buffering) (队列)	Simulate queue; supermarket checkout model	Bronze: Trace; Silver: Implement; Gold: Priority queue discussion
5	Formative: Data structures implementation test (30 marks)	Write pseudocode for stack/queue operations	All: Core operations; Gold: Error handling

Topic 4.3: Algorithms 算法 Suànfǎ

Week 14 — Searching Algorithms 搜索算法 Sōusuǒ Suànfǎ

Lesson	Content	Activities
1	Linear search: algorithm, trace, best/average/worst case (线性搜索)	Trace on small arrays; implement in pseudocode
2	Binary search: precondition, algorithm, trace; comparison to linear	Play "guess the number"; trace binary search tree
3	Algorithm efficiency: Big O notation introduction (HL depth)	Compare $O(n)$ vs $O(\log n)$ graphically
4	Trace tables: constructing and using to verify algorithm correctness (追踪表)	Complete trace tables for given algorithms
5	TOK: Is efficiency a property of an algorithm itself, or only of its implementation? Can we know an algorithm is optimal?	Discussion; link to P vs NP

TOK Knowledge Question: We say binary search is "more efficient" than linear search. But efficiency depends on sorted data. Is the claim "binary search is better" an absolute truth, or is it relative to assumptions about the world? How do hidden assumptions shape what we claim to know in computer science?

Week 15 — Sorting Algorithms 排序算法 Páixù Suànfǎ

Lesson	Content	Activities	Differentiation
1	Bubble sort: algorithm, trace, optimisation (flag), complexity (冒泡排序)	Sort cards manually; trace pseudocode	All: Trace; Gold: Prove $O(n^2)$ worst case
2	Selection sort: algorithm, trace, comparison to bubble	Implement; compare number of swaps	All: Implement; Gold: Analyse swap efficiency
3	Insertion sort: algorithm, trace; when it performs well	Sort a hand of cards; trace	All: Trace; Gold: Best case $O(n)$ analysis
4	Merge sort: divide and conquer, recursion preview (HL); quick sort mention	Trace merge sort; watch visualisation	Bronze: Understand divide/conquer; Silver: Trace; Gold: Implement recursively
5	Formative: Sorting algorithm comparison essay (8 marks)	Timed writing; peer assessment	All: Complete; Gold: Reference stability and adaptivity

Week 16 — Recursion 递归 Dìguī

Lesson	Content	Activities
1	Recursion concept: base case, recursive case, call stack (递归, 基本情况, 递归情况)	Factorial; trace call stack
2	Recursive algorithms: Fibonacci, binary search recursively	Implement; trace; compare iterative versions
3	Advantages and disadvantages of recursion; stack overflow risk	Debate: when to use recursion vs iteration
4	Backtracking introduction: maze solving, N-queens (HL enrichment)	Paper-based maze backtracking

Lesson	Content	Activities
5	IA Application: Identify algorithms needed for IA project	Algorithm design document

Topic 4.4: Software Development 软件开发 Ruǎnjiàn Kāifā

Week 17 — Testing and Debugging 测试与调试 Cèshì yǔ Tiádòushì

Lesson	Content	Activities
1	Testing types: black-box, white-box, unit, integration, system, acceptance (黑盒测试, 白盒测试)	Design test cases for a simple program
2	Test data: normal, boundary, erroneous/invalid (正常/边界/错误数据)	Generate test data sets; justify choices
3	Debugging techniques: trace tables, breakpoints, watch variables, rubber ducky	Debug intentionally buggy code
4	Documentation: internal comments, external user guide, technical manual	Write documentation for a given program
5	IA Application: Plan testing strategy for IA; draft test plan table	Test plan with normal, boundary, invalid cases

Week 18 — Topic 4 Review and Semester 2 Examination

Lesson	Content	Activities
1	Revision: Programming constructs and trace tables	Flashcard relay; trace table races
2	Revision: Sorting, searching, recursion	Algorithm card sort; complexity matching
3	Semester 2 Examination (Paper 2 mock, 1h30, SL/HL combined)	Formal conditions
4	Examination feedback; IA development check	DIRT; IA progress review
5	Semester roundup; Year 2 preview	Goal setting; celebrate IA milestones

Key Vocabulary (Topic 4)

English	中文	Pinyin	Definition
Pseudocode	伪代码	wěi dàimǎ	A structured, language-independent description of an algorithm
Variable	变量	biànlìang	A named storage location whose value can change
Array	数组	shùzǔ	A collection of elements of the same type, indexed by position

English	中文	Pinyin	Definition
Iteration	迭代/循环	diédài / xúnhuán	Repeating a set of instructions until a condition is met
Recursion	递归	dìguī	A function calling itself with a simpler version of the problem
Algorithm	算法	suànfǎ	A step-by-step procedure for solving a problem
Efficiency	效率	xiàolǜ	A measure of resource usage (time/space) by an algorithm
Trace table	追踪表	zhuīzōng biǎo	A table showing variable values at each step of an algorithm
Black-box testing	黑盒测试	hēi hé cèshì	Testing without knowledge of internal code structure
White-box testing	白盒测试	bái hé cèshì	Testing with knowledge of internal code structure

Teacher Checklist — End of Topic 4:

- ☐ All IB 4.1 – 4.4 syllabus points covered
- ☐ Students can write pseudocode fluently
- ☐ Students can construct and use trace tables
- ☐ Sorting and searching algorithms understood and compared
- ☐ IA test plans drafted
- ☐ TOK discussions documented
- ☐ Victoria Park Academy Shenzhen | IB DP Computer Science | Topic 4 Scheme of Work